

National Skill Development Mission, have made strides in manufacturing, there is a deeper need to expand into electronics, product design, and innovation, particularly beyond urban hubs.

“We are converting our best engineers into bankers and managers,” Kishor pointed out, questioning the cultural norm of IIT students preparing for MBAs by their third year. “One seat in IIT was lost for innovation.”

Students often pursue engineering because the focus is on securing a job, rather than building products or pursuing entrepreneurship. What is needed is a shift from the mindset of a job seeker to a job giver. A system where they will not be afraid of raising their hands and asking questions in class, guided by teachers who foster curiosity.

At the same time, established companies must also be held accountable, particularly when it comes to inclusion and innovation. While government-backed incubators can support startups in their early stages, it is the responsibility of larger enterprises to step forward and invest in talent. As Bhartendu rightly asked, “If you are taking subsidies, are you hiring for innovation or importing tech and rebranding it?”

India's future: from capable to world-class

“When I returned to India 20 years ago to start Nokia, we were terrified. But we went from zero to making seven phones a second, shipped to 110 countries, in just four years. At that time, Nokia's local material costs were lower than those of China. So when someone says it cannot be done, I say, we have already done it,” narrated Josh, citing his personal experience.

However, China has surged ahead in design. From chipset to embedded software, everything is designed and



T.S. Shankar, Sales Director, Analog Devices

built in-house. “That is the benchmark now,” he said. “Manufacturing is no longer enough. It is time for design to lead.”

India also faces a critical cultural point: the need to broaden participation and inclusion. “In our factory, 100% of the workforce was women. Maybe we need more women in hardware now,” Josh said. “Women made all the phones at Nokia and Foxconn. So why not have them lead in design too?”

Standing at an inflexion point, India's ESDM sector has demonstrated its ability to manufacture at scale. Now, it must prove it can design with vision. The country's future depends not just on building factories, but on building ecosystems that empower its millions of engineers to think independently, innovate fearlessly, and design proudly—for India and the world.

With tremendous engineering potential, India needs to remember what it is capable of. Like Hanuman from Ramayana was reminded by Jambavana, Josh put forward a powerful analogy from Indian mythology. The leap forward, as big as crossing oceans, will begin not with a new policy, but with the

realisation that it can do far more than it believes. And the time to leap is now.

Kishor Narang concluded with some clear imperatives: If we want to build a vibrant innovation and design ecosystem in the long run, then we need to focus on first building “the competence to innovate and design” for the next five to ten years. Suppose India wants to be a serious player in

this competition, while also fulfilling perennially urgent goals of boosting manufacturing and creating jobs. In that case, several fundamental changes in policy and mindset are necessary.

And for this, we have two crucial imperatives:

- Setting up a national institute of design and innovation in electronics, where the faculty shall not be merely academicians and researchers but rather practicing design engineers from respective domains in the industry.

- Establishing large, independent design houses with the capability to design end-to-end, high-end products ready for mass manufacture and transferring the complete know-how to Indian industries willing to take up manufacturing. **EFY**

The article is based on insights from the panel discussion titled “Striding Towards Global Leadership in ESDM: Role and Strategies for India's Ecosystem,” held on 28 February 2025 during the smartBHARAT summit at India Electronics Week in Bengaluru. It was moderated by N. Kishor Narang, Mentor and Principal Design Architect, Narnix Technolabs Pvt Ltd, with panellists Jaswinder Ahuja, ex-Corporate Vice President of Cadence Design Systems; Josh Foulger, President, Electronics, Zetwerk Manufacturing; T.S. Shankar, Sales Director, Analog Devices; and Bhartendu Mishra, Managing Director, Arrow Electronics, India.